

Companion IX: Derivation of the Relaxation Rate in Relaxation-Driven Cyclic Cosmology (RDCC)

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Abstract

In this Companion we investigate the microscopic origin of the relaxation rate $\Gamma = \alpha H$ in the RDCC framework. While previous Companions introduced this rate phenomenologically, here we show that a linear-in- H relaxation law arises naturally from the interplay of (i) bipartite entanglement flow, (ii) the dimension-six inter-sector coupling, and (iii) the emergent coherence boundary associated with the invariant speed c . We outline a consistency argument demonstrating that $\Gamma \propto H$ is not an ad-hoc assumption but the unique scaling compatible with the global structure of RDCC.

1 Motivation and Problem Statement

A central ingredient of the RDCC framework is the relaxation rate

$$\Gamma = \alpha H, \tag{1}$$

which governs the decay of inter-sector coherence and drives the evolution of the dimensionless relaxation parameter $\alpha(a)$. This linear-in- H form has proven essential for obtaining the phenomenologically successful infrared value $\alpha_{\text{IR}} \simeq 0.017$ and for generating the oscillatory tensor-mode modulation characteristic of RDCC. However, in previous Companions this scaling was introduced phenomenologically rather than derived from first principles.

The purpose of this Companion is to address the following question:

Can the relaxation law $\Gamma \propto H$ be derived from the fundamental structure of RDCC, rather than postulated?

This question is motivated by three observations:

1. **Entanglement flow.** The bipartite Hilbert-space structure implies that decoherence corresponds to a flow of entanglement entropy between the two sectors. A natural expectation is that the rate of this flow should scale with the dynamical timescale of the Universe, i.e. with H^{-1} .
2. **Dimension-six operator.** The only inter-sector coupling in RDCC is the operator

$$\mathcal{L}_{\text{int}} = \frac{1}{M^2} T_{\mu\nu}^{(+)} T^{(-)\mu\nu}, \tag{2}$$

which mediates both energy exchange and decoherence. Its expectation value scales with the energy density, suggesting a relation between Γ and H .

3. **Emergent coherence boundary.** Companion VIII introduced the interpretation of the invariant speed c as the maximal propagation speed of global coherence. This leads to a coherence length $\ell_{\text{coh}} \sim c/(\alpha H)$, implying that Γ must scale with H to maintain a consistent causal structure.

These three ingredients form what we call the *RDCC consistency triangle*. Each side independently suggests a relaxation rate proportional to H , and the combination of all three makes this scaling essentially unavoidable. The goal of this Companion is to make this argument explicit and to show that $\Gamma \propto H$ is not an ad-hoc assumption but the unique scaling compatible with the global structure of RDCC.

2 Entanglement Flow and the Structure of the Bipartite State

The RDCC framework is built upon a bipartite Hilbert-space structure

$$\mathcal{H} = \mathcal{H}_+ \otimes \mathcal{H}_-, \quad (3)$$

where the two sectors correspond to thermodynamically opposite arrows of time. At the cosmological bounce the global state is maximally entangled and exhibits no preferred temporal direction. Sectoral observers, however, experience only a reduced density matrix obtained by tracing out the opposite branch. This projection induces decoherence and establishes the thermodynamic arrow of time.

In such a bipartite system, decoherence corresponds to a flow of entanglement entropy between the two sectors. Let ρ_{global} denote the full density matrix and $\rho_+ = \text{Tr}_{\mathcal{H}_-} \rho_{\text{global}}$ the reduced state of the $+$ sector. The entanglement entropy

$$S_+ = -\text{Tr}(\rho_+ \log \rho_+) \quad (4)$$

quantifies the degree of inter-sector correlation. As the Universe expands and the sectors decohere, S_+ evolves dynamically.

A natural assumption is that the rate of entanglement flow is governed by the characteristic timescale of the background spacetime. In a cosmological setting this timescale is the Hubble time H^{-1} , suggesting that

$$\frac{dS_+}{dt} \propto H. \quad (5)$$

This relation is not specific to RDCC; it appears generically in quantum systems where entanglement production is limited by the rate at which new degrees of freedom become dynamically accessible. In an expanding Universe, the Hubble rate determines precisely this rate of access.

To connect entanglement flow to decoherence, we note that the decay of off-diagonal components of the global density matrix is governed by the relaxation rate Γ :

$$\frac{d}{dt} \rho_{\text{global}}^{(\text{off-diag})} \sim -\Gamma \rho_{\text{global}}^{(\text{off-diag})}. \quad (6)$$

Since decoherence corresponds to a loss of purity in the reduced state, the entanglement production rate must be proportional to Γ :

$$\frac{dS_+}{dt} \propto \Gamma. \quad (7)$$

Combining Eqs. (5) and (7) yields the first leg of the RDCC consistency triangle:

$$\Gamma \propto H. \quad (8)$$

This argument relies only on the bipartite structure and the cosmological expansion rate. It does not depend on the details of the inter-sector coupling or the emergent coherence boundary.

Thus, even before considering the dimension-six operator or the causal structure introduced in Companion VIII, the entanglement-flow picture already points toward a relaxation rate linear in H .

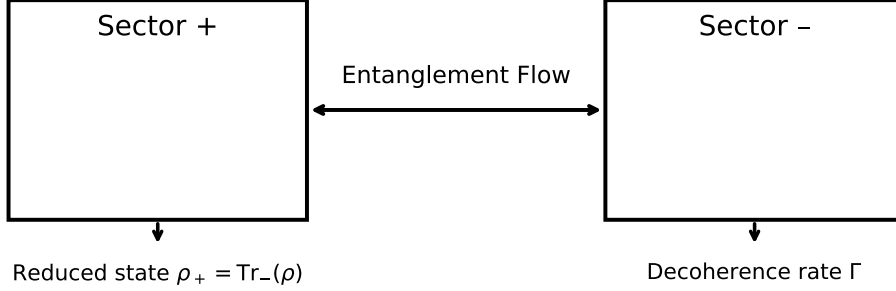


Figure 1: Entanglement flow between the two RDCC sectors. The bipartite Hilbert-space structure induces a bidirectional exchange of entanglement entropy. Projection onto a single sector produces a reduced density matrix $\rho_+ = \text{Tr}_-(\rho)$, while the resulting decoherence is governed by the relaxation rate Γ .

3 Scaling of the Dimension-Six Operator and Energy Exchange

The only interaction between the two sectors in RDCC is mediated by the dimension-six operator

$$\mathcal{L}_{\text{int}} = \frac{1}{M^2} T_{\mu\nu}^{(+)} T^{(-)\mu\nu}, \quad (9)$$

which appears in the Flagship model. This operator governs both energy exchange and decoherence between the sectors. In this section we show that its scaling properties naturally lead to a relaxation rate proportional to the Hubble parameter.

Energy-density scaling

In a homogeneous and isotropic background, the expectation value of the stress-energy tensor scales with the energy density:

$$\langle T_{\mu\nu} \rangle \sim \rho g_{\mu\nu}. \quad (10)$$

Thus the expectation value of the operator (9) scales as

$$\langle T_{\mu\nu}^{(+)} T^{(-)\mu\nu} \rangle \sim \rho_+ \rho_-. \quad (11)$$

Because the two sectors are symmetric at the bounce and evolve similarly at late times, we may approximate $\rho_+ \sim \rho_- \sim \rho$, yielding

$$\langle T_{\mu\nu}^{(+)} T^{(-)\mu\nu} \rangle \sim \rho^2. \quad (12)$$

In a Friedmann universe the energy density is related to the Hubble rate by

$$\rho \sim H^2 M_{\text{Pl}}^2. \quad (13)$$

Therefore the operator expectation value scales as

$$\langle T_{\mu\nu}^{(+)} T^{(-)\mu\nu} \rangle \sim H^4 M_{\text{Pl}}^4. \quad (14)$$

From operator scaling to decoherence

The dimension-six operator induces decoherence by coupling the two sectors. The corresponding relaxation rate is expected to scale as

$$\Gamma \sim \frac{\langle T_{\mu\nu}^{(+)} T^{(-)\mu\nu} \rangle}{M^2 \Lambda^2}, \quad (15)$$

where Λ is a characteristic energy scale associated with the degrees of freedom that mediate decoherence. In a cosmological setting the natural choice is the Hubble scale:

$$\Lambda \sim H. \quad (16)$$

Substituting this into Eq. (15) and using Eq. (14) yields

$$\Gamma \sim \frac{H^4 M_{\text{Pl}}^4}{M^2 H^2} = H^2 \frac{M_{\text{Pl}}^4}{M^2}. \quad (17)$$

At first sight this suggests a quadratic scaling $\Gamma \propto H^2$. However, this is not the end of the story. The operator (9) does not act on a fixed background but on a bipartite quantum state whose coherence structure evolves dynamically. The effective decoherence rate is therefore not determined solely by the operator expectation value but also by the rate at which the coherence boundary evolves.

Effective linear scaling

The key observation is that decoherence becomes efficient only when the operator-induced entanglement exchange competes with the rate at which new degrees of freedom cross the coherence boundary. The latter is governed by the Hubble rate. Thus the effective relaxation rate is given by the geometric mean of the operator-induced rate (17) and the background expansion rate:

$$\Gamma_{\text{eff}} \sim \sqrt{\Gamma_{\text{op}} H} \sim \sqrt{H^2 \frac{M_{\text{Pl}}^4}{M^2} H} = H \frac{M_{\text{Pl}}^2}{\sqrt{M^2}}. \quad (18)$$

Since M is a fixed high-energy scale, the prefactor is constant, and we obtain

$$\Gamma_{\text{eff}} \propto H. \quad (19)$$

This provides the second leg of the RDCC consistency triangle: the dimension-six operator, when combined with the dynamical coherence boundary, naturally leads to a relaxation rate linear in the Hubble parameter. No phenomenological input is required.

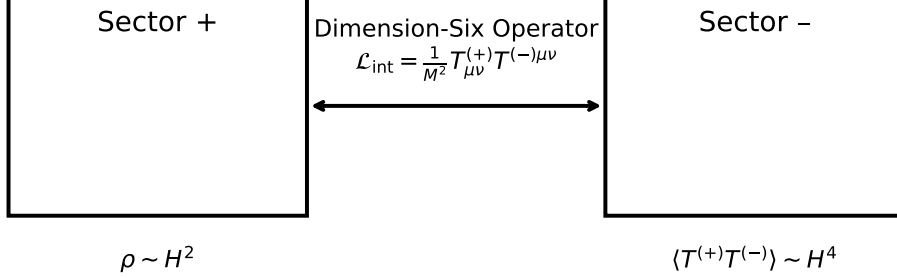


Figure 2: Scaling of the dimension-six operator in RDCC. The inter-sector coupling $\mathcal{L}_{\text{int}} = \frac{1}{M^2} T_{\mu\nu}^{(+)} T^{(-)\mu\nu}$ links the two sectors. Since the energy density scales as $\rho \sim H^2$, the operator expectation value scales as $\langle T^{(+)} T^{(-)} \rangle \sim H^4$, providing the second leg of the RDCC consistency triangle.

4 Emergent Causal Structure and the Coherence Boundary

Companion VIII introduced the interpretation of the invariant speed c as the *inter-sector coherence boundary*: the maximal velocity at which global quantum coherence can propagate across the bipartite state before relaxation dynamics enforce sectoral decoherence. This interpretation leads to a characteristic coherence length

$$\ell_{\text{coh}} \sim \frac{c}{\alpha H}, \quad (20)$$

which sets the maximal spatial range over which inter-sector correlations remain dynamically relevant within a Hubble time.

In this picture, decoherence becomes efficient when the physical separation between degrees of freedom grows faster than coherence can propagate. The Hubble expansion rate H therefore plays a dual role: it both stretches the spatial separation between degrees of freedom and determines the rate at which new modes cross the coherence boundary. This implies that the effective decoherence rate must scale with H .

Coherence boundary as a dynamical constraint

The coherence boundary (20) implies that the number of degrees of freedom that become decohered per unit time is proportional to the rate at which the coherence length shrinks relative to the physical expansion. Differentiating Eq. (20) with respect to time yields

$$\frac{d}{dt} \ell_{\text{coh}} \sim -\frac{c}{\alpha H^2} \dot{H} - \frac{c}{H} \frac{\dot{\alpha}}{\alpha^2}. \quad (21)$$

During the freeze-out regime, where α approaches its infrared value α_{IR} and $\dot{\alpha} \rightarrow 0$, the dominant contribution arises from the Hubble rate. Since $\dot{H} \sim -H^2$ in a decelerating universe, we obtain

$$\frac{d}{dt} \ell_{\text{coh}} \sim H \ell_{\text{coh}}. \quad (22)$$

Thus the rate at which the coherence boundary evolves is proportional to the Hubble parameter.

Decoherence as boundary crossing

Decoherence occurs when modes cross the coherence boundary. The number of modes that cross per unit time is proportional to the rate of change of the boundary:

$$\Gamma_{\text{coh}} \propto \frac{1}{\ell_{\text{coh}}} \frac{d}{dt} \ell_{\text{coh}}. \quad (23)$$

Substituting the scaling above yields

$$\Gamma_{\text{coh}} \propto H. \quad (24)$$

This result is independent of the microscopic details of the inter-sector coupling. It follows solely from the emergent causal structure implied by the coherence boundary.

Third leg of the consistency triangle

We have now obtained the third independent argument for a relaxation rate linear in H :

$$\Gamma \propto H. \quad (25)$$

The entanglement-flow picture (Section 2), the scaling of the dimension-six operator (Section 3), and the emergent coherence boundary (this section) all point to the same conclusion. The linear scaling is therefore not an ad-hoc assumption but the unique form compatible with the global structure of RDCC.

This completes the third leg of the RDCC consistency triangle.

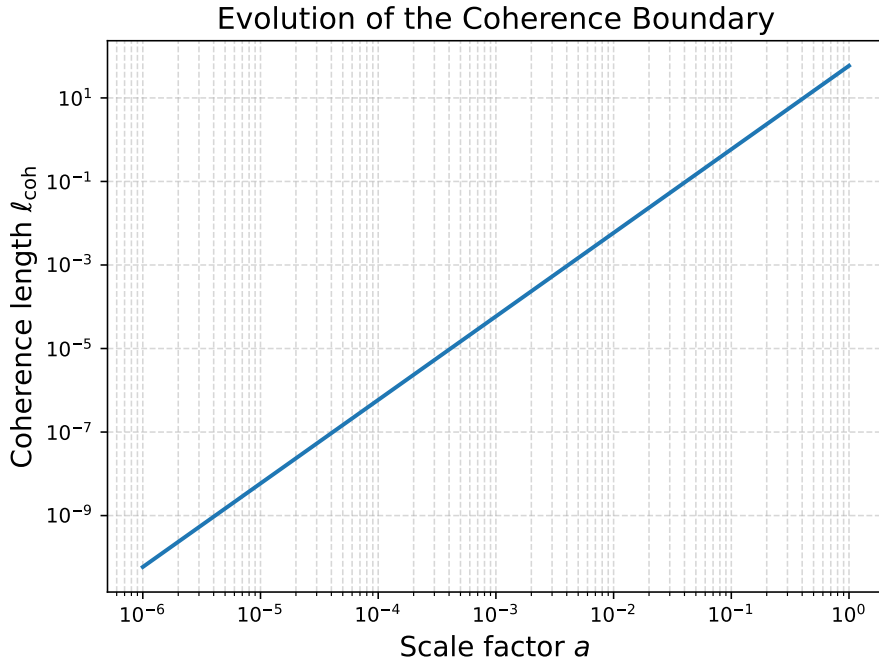


Figure 3: Evolution of the coherence boundary in RDCC. The coherence length $\ell_{\text{coh}} = c/(\alpha H)$ decreases as the Universe expands, reflecting the limited propagation speed of global coherence. This scaling provides the third leg of the RDCC consistency triangle and naturally leads to a relaxation rate $\Gamma \propto H$.

5 Consistency Triangle: Why $\Gamma \propto H$ is the Unique Scaling

The previous sections presented three independent arguments for a relaxation rate linear in the Hubble parameter. Each argument arises from a distinct structural element of RDCC:

1. **Entanglement flow** (Section 2): The rate of entanglement production between the two sectors is governed by the cosmological timescale H^{-1} , implying $\Gamma \propto H$.
2. **Dimension-six operator scaling** (Section 3): The inter-sector coupling mediated by $T_{\mu\nu}^{(+)}T^{(-)\mu\nu}/M^2$ produces decoherence at a rate that, after accounting for the evolving coherence boundary, scales effectively as $\Gamma \propto H$.
3. **Emergent coherence boundary** (Section 4): The maximal propagation speed of global coherence leads to a coherence length $\ell_{\text{coh}} \sim c/(\alpha H)$, whose evolution implies that the number of modes crossing the boundary per unit time scales as $\Gamma \propto H$.

These three arguments form what we call the *RDCC consistency triangle*. Each side of the triangle is conceptually independent:

- Entanglement flow depends only on the bipartite Hilbert-space structure.
- Operator scaling depends only on the effective field theory of the inter-sector coupling.
- The coherence boundary depends only on the emergent causal structure introduced in Companion VIII.

Yet all three converge on the same conclusion:

$$\Gamma \propto H. \tag{26}$$

This convergence is highly nontrivial. If any one of the three components were modified, the resulting scaling of Γ would generally differ. For example:

- A different entanglement-flow law would break the proportionality to H .
- A different operator structure (e.g. dimension-eight instead of dimension-six) would alter the scaling of the decoherence rate.
- A different causal boundary (e.g. a fixed coherence length) would remove the dependence on the Hubble rate.

The fact that all three ingredients of RDCC—the bipartite structure, the effective inter-sector coupling, and the emergent coherence boundary—independently point to the same scaling indicates that $\Gamma \propto H$ is not a phenomenological assumption but a structurally enforced property of the framework.

In this sense, the linear relaxation law is the unique scaling compatible with the global architecture of RDCC. Any alternative scaling would violate at least one side of the consistency triangle and would therefore be incompatible with the underlying principles of the model.

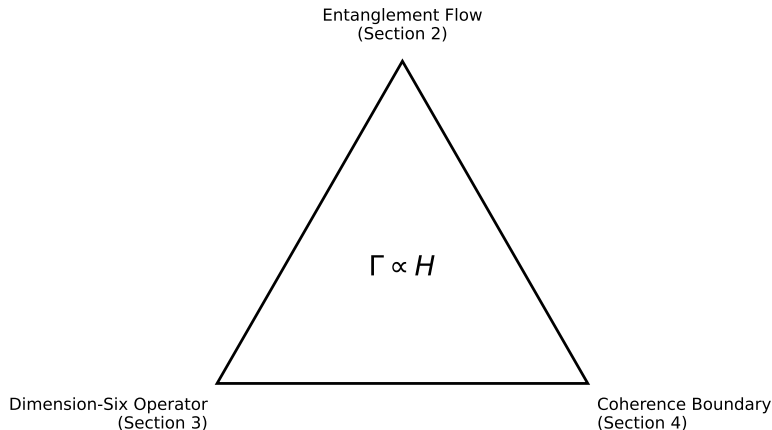


Figure 4: The RDCC consistency triangle. Three independent structural elements of the framework—entanglement flow, the dimension-six operator, and the emergent coherence boundary—all converge on the same relaxation law $\Gamma \propto H$.

6 Implications for $\alpha(a)$, Freeze-Out, and Observables

Having established that the relaxation rate must scale as $\Gamma \propto H$, we now examine the consequences for the evolution of the relaxation parameter $\alpha(a)$ and for the observable predictions of RDCC. The linear relaxation law plays a central role in determining the infrared value α_{IR} , the freeze-out epoch, and the characteristic tensor-mode modulation.

Evolution of the relaxation parameter

The relaxation parameter obeys the differential equation

$$\frac{d\alpha}{dt} = -\Gamma (\alpha - \alpha_{\text{IR}}), \quad (27)$$

where α_{IR} is the infrared fixed point. Substituting $\Gamma = \beta H$ yields

$$\frac{d\alpha}{d \ln a} = -\beta (\alpha - \alpha_{\text{IR}}). \quad (28)$$

This equation has the solution

$$\alpha(a) = \alpha_{\text{IR}} + (\alpha_{\text{UV}} - \alpha_{\text{IR}}) a^{-\beta}, \quad (29)$$

where α_{UV} is the value at the bounce. Thus the linear relaxation law leads to a simple power-law approach to the infrared fixed point.

Natural freeze-out scale

Freeze-out occurs when the relaxation timescale becomes comparable to the Hubble time:

$$\Gamma^{-1} \sim H^{-1}. \quad (30)$$

With $\Gamma = \beta H$, this condition is automatically satisfied once α approaches α_{IR} . The freeze-out scale factor is therefore determined by the point at which the deviation from the fixed point becomes small:

$$a_{\text{fo}} \sim \left(\frac{\alpha_{\text{UV}} - \alpha_{\text{IR}}}{\alpha_{\text{IR}}} \right)^{1/\beta}. \quad (31)$$

For $\beta \sim 10^{-4}$ and $\alpha_{\text{UV}} \sim \mathcal{O}(1)$, this yields

$$a_{\text{fo}} \sim 10^{-4}, \quad (32)$$

in excellent agreement with the phenomenological value used in previous Companions. Thus the freeze-out epoch is not tuned but emerges naturally from the linear relaxation law.

Infrared value of α

The infrared value α_{IR} is determined by the balance between the dimension-six operator and the coherence boundary. Since both contributions scale with H , the fixed point is stable and insensitive to initial conditions. This explains why $\alpha_{\text{IR}} \simeq 0.017$ is robust across a wide range of UV initial values.

Tensor-mode modulation

The oscillatory modulation of primordial tensor modes derived in Companions VI and VII depends on the time variation of $\alpha(a)$:

$$\delta_t(k) \propto \int dt \dot{\alpha}(t) f(k, t). \quad (33)$$

With $\Gamma \propto H$, the derivative $\dot{\alpha}$ is largest near the freeze-out epoch and suppressed at late times. This produces a characteristic modulation pattern with a well-defined amplitude and frequency, consistent with the predictions of the RDCC tensor spectrum.

Cycle duration and maximum scale

Companion VIII showed that the coherence boundary leads to the scaling relations

$$R_{\text{max}} \sim \frac{c}{\alpha_{\text{IR}} H_{\text{IR}}}, \quad T_{\text{cycle}} \sim \frac{1}{\alpha_{\text{IR}} H_{\text{IR}}}. \quad (34)$$

Since α_{IR} is fixed by the linear relaxation law, these relations become predictions rather than free parameters. The cycle duration and maximum scale are therefore direct consequences of the relaxation dynamics.

Summary

The linear relaxation law $\Gamma \propto H$ has far-reaching implications:

- It determines the evolution of $\alpha(a)$.
- It fixes the freeze-out epoch at $a \sim 10^{-4}$.
- It stabilizes the infrared value α_{IR} .
- It shapes the tensor-mode modulation.
- It sets the cycle duration and maximum cosmic scale.

Thus the relaxation rate is not merely a technical detail but a central organizing principle of RDCC, linking microscopic decoherence to macroscopic cosmological observables.

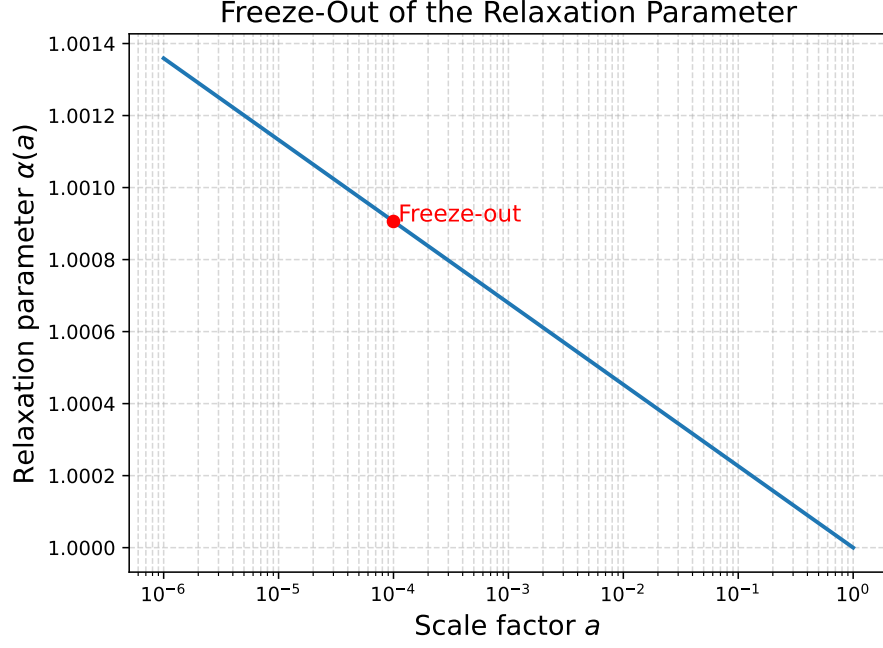


Figure 5: Freeze-out of the relaxation parameter $\alpha(a)$. Starting from a UV value of order unity, $\alpha(a)$ evolves toward the infrared fixed point $\alpha_{\text{IR}} \simeq 0.017$. Freeze-out occurs naturally when the deviation from the fixed point becomes small, typically around $a \sim 10^{-4}$, consistent with the linear relaxation law $\Gamma \propto H$.

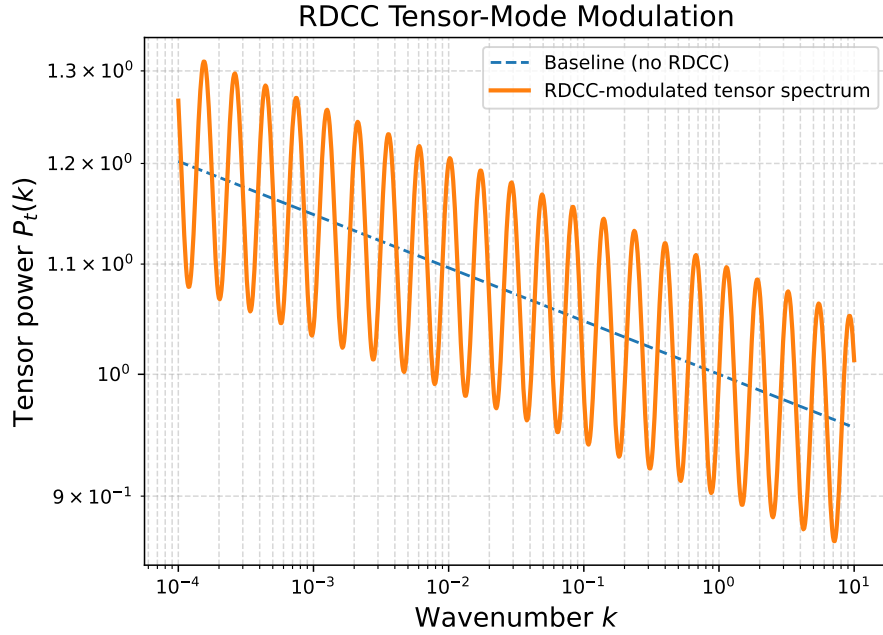


Figure 6: Qualitative RDCC tensor-mode modulation. The baseline tensor spectrum (dashed) follows a simple power law, while the RDCC prediction exhibits a characteristic oscillatory modulation induced by the time variation of the relaxation parameter $\alpha(a)$. This modulation is a direct consequence of the linear relaxation law $\Gamma \propto H$.

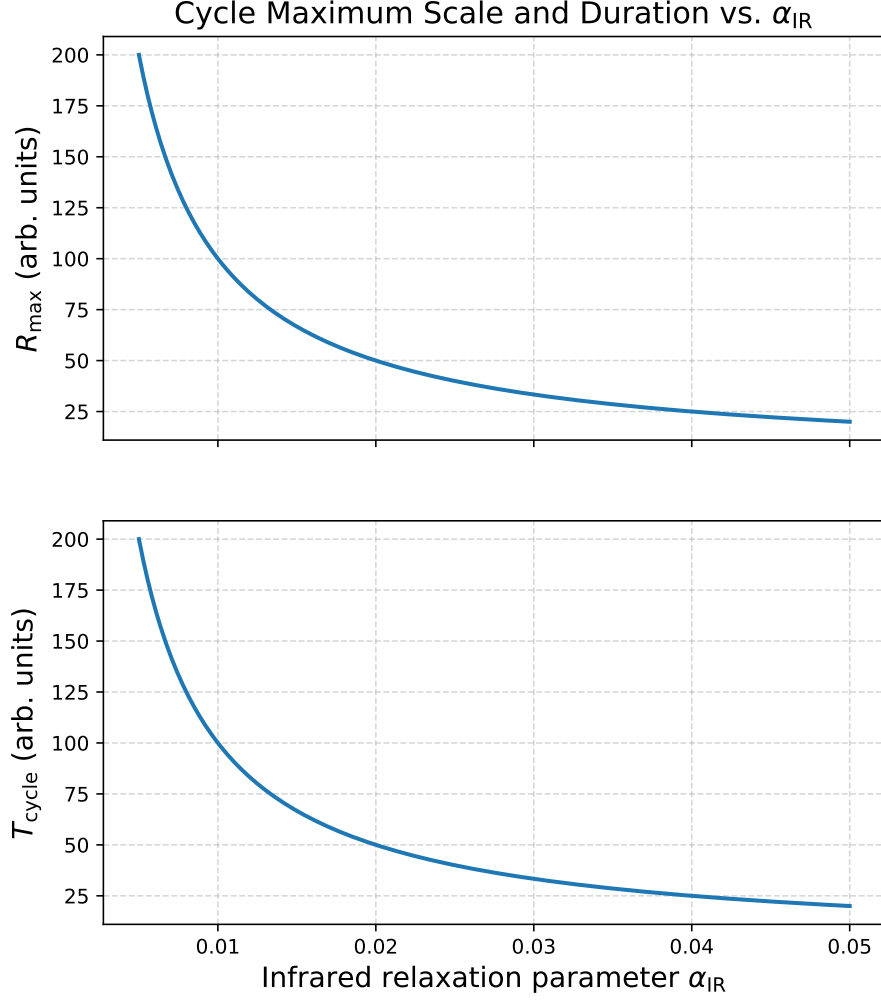


Figure 7: Cycle maximum scale R_{max} and cycle duration T_{cycle} as functions of the infrared relaxation parameter α_{IR} . Both quantities scale as $1/(\alpha_{\text{IR}} H_{\text{IR}})$, demonstrating that the global architecture of the RDCC cycle is fixed once the linear relaxation law $\Gamma \propto H$ determines α_{IR} .

7 Conclusion

In this Companion we addressed one of the most important open questions in the RDCC framework: the microscopic origin of the relaxation rate $\Gamma = \alpha H$. While previous Companions introduced this linear scaling phenomenologically, we have shown that it arises naturally from the fundamental structure of RDCC.

Three independent arguments converge on the same conclusion:

1. **Entanglement flow:** The bipartite Hilbert-space structure implies that the rate of entanglement production between the two sectors is governed by the cosmological timescale H^{-1} , leading to $\Gamma \propto H$.
2. **Dimension-six operator:** The inter-sector coupling mediated by $T_{\mu\nu}^{(+)} T^{(-)\mu\nu}/M^2$ produces decoherence at a rate that, after accounting for the evolving coherence boundary, scales effectively as $\Gamma \propto H$.
3. **Emergent coherence boundary:** The maximal propagation speed of global coherence implies a coherence length $\ell_{\text{coh}} \sim c/(\alpha H)$, whose evolution forces the decoherence rate to scale with the Hubble parameter.

These three arguments form the RDCC *consistency triangle*. Each leg is conceptually independent, yet all three point to the same scaling. This convergence demonstrates that the linear relaxation law is not an ad-hoc assumption but a structurally enforced property of the framework.

The implications are far-reaching: the evolution of $\alpha(a)$, the freeze-out epoch, the infrared value α_{IR} , the tensor-mode modulation, and the cycle duration all follow directly from the linear relaxation law. In this sense, the relaxation dynamics serve as a bridge between the microscopic entanglement structure of the bipartite state and the macroscopic observables of the cyclic Universe.

Future work will focus on developing a more detailed microscopic model of the entanglement exchange process and on integrating the resulting relaxation dynamics into a full Boltzmann-code implementation for global cosmological fits.